

Shared burst dynamics and limits of neuron-specific identifiability in sparse-positive zebrafish fluorescence imaging

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Abstract

Sparse fluorescence recordings pose a measurement problem before they pose a learning problem: event times may be partially known, active neurons may be incompletely annotated, and nearby tissue can express the same recording-level dynamics as a labeled region of interest. We present an identifiability-first analysis framework for a zebrafish fluorescence recording containing 4[†] shared burst intervals and 79[†] labeled neuron–burst occurrences distributed over 27[†] original spatial sites. The framework separates immutable observation-site geometry from provisional neuron identity, exports native-amplitude traces from Raw and processed representations, characterizes acquisition noise and common-mode structure, and evaluates temporal and spatial specificity with block-shift and spatial-shift controls. Preliminary repository evidence indicates that a two-frame independent component is nearly equivalent to signed temporal differentiation, while Raw event-aligned traces exhibit a reproducible population waveform that is also present in adjacent tissue. Local ROI-minus-annulus residual amplitude increases during labeled events, but a universal neuron-specific residual waveform has not yet been established. The confirmatory study therefore tests whether amplitude, signal-to-noise ratio, timing, spatial compactness, and known-positive recoverability are stable neuron- or site-associated measurement phenotypes. Precision is estimated only within an exhaustively reviewed bounded field; elsewhere, evaluation is

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restricted to positive-unlabeled quantities. This design reframes unsupervised neuron detection as a sequence of testable measurement and identifiability questions and defines the additional evidence required for stronger biological or detection claims.

Keywords: zebrafish, calcium imaging, sparse-positive annotation, source identifiability, fluorescence traces, positive-unlabeled evaluation, independent component analysis

Working manuscript skeleton. Daggered values and figures are provisional outputs from the pre-repair repository. Author, ethics, funding, data-release, and final adjudication fields require review. The release build must disable provisional mode and contain no red placeholders.

1. Introduction

Optical recordings can expose the coordinated activity of large neuronal populations, including whole-brain dynamics in larval zebrafish (Ahrens et al., 2013). Their apparent scale, however, does not remove a basic measurement problem: a fluorescent transient observed at a labeled location may reflect a neuron, nearby neuropil, motion, shared illumination or acquisition effects, or a mixture of these sources. Mature calcium-imaging workflows therefore separate registration, spatial source extraction, background modeling, temporal trace estimation, and event inference rather than treating them as one interchangeable detection problem (Pnevmatikakis et al., 2016; Giovannucci et al., 2019; Zhou et al., 2018; Keemink et al., 2018; Friedrich et al., 2017).

The present study occupies a data regime in which conventional supervised-learning claims are especially difficult to support. The available canonical recording contains a small number of shared burst intervals, and the positive annotations identify only a subset of plausible neurons. The labels are useful anchors but do not define an exhaustive negative class. Consequently, an unmatched algorithmic candidate cannot automatically be called a false positive, and the absence of a label cannot be interpreted as biological inactivity. This is a positive-unlabeled setting in which the labeling mechanism itself may favor bright, isolated, or otherwise readily visible neurons (Bekker and Davis, 2020).

These constraints motivate an identifiability-first question. Rather than asking only whether an operator assigns a high score at a labeled location, we

ask which components of the observed signal are temporally exceptional, spatially localized, reproducible within a site, shared across the field, and stable across repeated bursts. This distinction is essential because a representation can be useful for detecting change while discarding fluorescence amplitude, distorting peak time, or retaining the same waveform in surrounding tissue. Likewise, a high event-aligned correlation can reflect a recording-level burst template without establishing a neuron-specific activation signature.

The repository already contains evidence that illustrates this distinction. A learned two-frame independent component is nearly collinear with signed temporal difference; event-aligned Raw traces contain a strong shared waveform; and local background subtraction increases event-associated residual amplitude without yet establishing a universal residual morphology. These results are informative boundary conditions, but the current identity handling, sparse labels, and reuse of the same four bursts require a protected confirmatory analysis before final numerical claims are made.

We therefore develop a reproducible analysis program with four objectives. First, we separate immutable observation-site geometry from provisional canonical-neuron identity and export a complete Raw-to-pipeline trace atlas. Second, we characterize acquisition noise, common-mode activity, and spatial contamination before interpreting neuron-level variation. Third, we partition scalar and functional variability into site-, burst-, and residual components and test whether amplitude, signal-to-noise ratio, timing, and spatial compactness define stable measurement phenotypes. Fourth, we evaluate unsupervised representations using positive-unlabeled-compatible metrics and reserve precision estimation for a bounded field that receives exhaustive blinded review. The intended contribution is not a claim that a single operator solves neuron detection, but a framework for determining what the available recording can and cannot identify.

2. Materials and methods

2.1. Study design and inferential scope

The study is an intensive methodological analysis of one canonical zebrafish fluorescence recording. The current repository artifacts describe 4[†] shared burst intervals and 79[†] labeled neuron–burst occurrences. These occurrences arise from 27[†] original spatial observation sites and 26[†] proposed canonical neuron identities. The recording, not the individual frame or pixel, is the highest independent experimental unit. All population-level statements are

therefore excluded; inferential claims concern temporal and spatial structure within this recording.

Existing numerical results are treated as discovery-stage evidence. The confirmatory run freezes the label contract, candidate representations, spatial supports, temporal windows, null procedures, and primary estimands before regenerating the manuscript outputs. Analyses are organized in versioned stages, each of which emits a machine-readable metric file, a validation report, an artifact index, and a source map.

2.2. Observation, site, and canonical-identity contract

Each labeled occurrence is represented by an immutable record

$$o_i = (q_i, s_i, c_i, b_i, x_i, y_i, t_i^{\text{start}}, t_i^{\text{stop}}, \ell_i), \quad (1)$$

where q_i is a unique observation identifier, s_i is an immutable spatial-site identifier, c_i is a provisional canonical-neuron identifier, b_i is the burst, (x_i, y_i) is the source coordinate or mask reference, and ℓ_i stores annotation provenance and confidence. Trace extraction is keyed by s_i and geometry, never by c_i alone. A canonical merge is an aggregation hypothesis applied after extraction; it cannot overwrite or redefine an original spatial site.

This distinction is required because the current repository provisionally maps original sites ROI 010 and ROI 015 to one canonical identity although their coordinates differ. The protected rerun preserves both spatial traces, generates a direct comparison packet, and reports site-level results regardless of the eventual merge decision. Canonical-neuron summaries are generated only for identity decisions that have been explicitly adjudicated or are accompanied by an unresolved-status sensitivity analysis.

Frame intervals shown to reviewers are one-based and inclusive. Internal array slices are zero-based and half-open. Coordinates follow x = column and y = row. Automated validation checks bounds, interval conversion, duplicate observation identifiers, coordinate collisions, and one-to-one trace provenance.

2.3. Annotation contract and timing uncertainty

Annotations are interpreted as sparse positives rather than exhaustive ground truth. Each occurrence stores neuron-presence confidence, activity confidence, timing confidence, morphology notes, reviewer state, and provenance. Original annotations remain immutable. Suggested revisions are written to a separate adjudication table and never silently replace the source labels.

Burst 2 is prespecified as timing-uncertain because preliminary traces suggest a weaker event and a pipeline peak shift opposite to that observed in the other bursts. Automated onset, peak, and offset proposals are generated from Raw, annulus, residual, difference, and frozen-ICA traces. The primary analysis is reported using the original timing, with sensitivity analyses using the approved revision and excluding burst 2 entirely. Human approval is required before an annotation is changed, but unresolved timing does not block analyses that remain valid under the original contract.

2.4. Trace extraction and atlas

For every occurrence, the pipeline exports synchronized full-context and event-centered traces from the following representations:

- (a) Raw ROI mean;
- (b) baseline-corrected Raw and, when physically appropriate, $\Delta F/F_0$;
- (c) robustly standardized Raw;
- (d) local annulus and outer-ring means;
- (e) amplitude-preserving ROI-minus-annulus residual;
- (f) signed adjacent-frame difference;
- (g) the frozen two-frame ICA component;
- (h) selected spatial-context scores, including local coherence and lagged recurrence.

Primary traces preserve native timing and amplitude. Shape-only displays that peak-align or scale each event are generated separately and labeled as such. Saturated samples, field-boundary membership, motion quality, and annotation timing are overlaid on the atlas.

For a trace $Y_i(t)$ and pre-event baseline window W_i^{pre} , the robust baseline and scale are

$$\tilde{Y}_i = \text{median}_{t \in W_i^{\text{pre}}} Y_i(t), \quad (2)$$

$$\hat{\sigma}_i = 1.4826 \text{ MAD}_{t \in W_i^{\text{pre}}} Y_i(t). \quad (3)$$

The native-amplitude event peak is

$$A_i = \max_{t \in W_i^{\text{event}}} \{Y_i(t) - \tilde{Y}_i\}, \quad (4)$$

and robust peak signal-to-noise ratio is $A_i/(\hat{\sigma}_i + \epsilon)$. Additional metrics include signed minimum and maximum, area above baseline, rise time, time to peak, decay half-time, full width at half maximum, baseline slope, saturation fraction, Raw-pipeline peak-time offset, Raw-pipeline correlation, annulus coupling, residual amplitude, and spatial compactness. Metrics with unstable denominators or censored peaks are flagged rather than imputed silently.

2.5. Acquisition and noise characterization

Acquisition properties are estimated before biological interpretation. Quiet adjacent-frame differences are used to assess an intensity-dependent Poisson-Gaussian model (Foi et al., 2008):

$$\text{Var}(Y_{t+1} - Y_t \mid \bar{Y}_t) \approx 2 \left(\sigma_r^2 + \alpha \bar{Y}_t \right), \quad (5)$$

where σ_r^2 represents an intensity-independent component and α an intensity-dependent component. Robust binned regression reports the slope, intercept, goodness of fit, and departures associated with saturation.

Temporal acquisition diagnostics include autocorrelation, power spectral density, baseline drift, photobleaching trend, frame-difference variance, and change-point sensitivity. Spatial diagnostics include saturation maps, row/column discontinuities, local covariance, effective rank, motion residuals, and field-specific intensity statistics. Other available recordings are not treated as biological validation. They form an unlabeled acquisition-QC cohort used to characterize failure regimes and to test whether the canonical recording is unusual in noise, motion, saturation, drift, coherence, or transient density.

2.6. Temporal controls and operator identification

Temporal event evidence is tested against block-matched circular or admissible shifted windows that preserve event duration, recording region, and temporal dependence. Shifts are excluded when they overlap any labeled event or protected buffer. The resampling unit is the burst or observation site as appropriate; individual frames are not treated as independent replicates.

For the learned two-frame component $z_t = w_0 Y_t + w_1 Y_{t+1}$, operator identity is assessed by cosine similarity, Pearson and Spearman correlation, coefficient of determination, and top-activation overlap against interpretable two-frame operators. The main comparator is signed difference $D_t = Y_{t+1} - Y_t$. A near-equivalence result is interpreted as change detection rather than recovery of an independent neuronal source. Detection utility and trace fidelity are evaluated separately.

2.7. Spatial specificity and matched spatial nulls

For each event, an event-minus-baseline map is defined as

$$D_i(x, y) = \frac{1}{|W_i^{\text{event}}|} \sum_{t \in W_i^{\text{event}}} Y(x, y, t) - \text{median}_{t \in W_i^{\text{pre}}} Y(x, y, t). \quad (6)$$

Radial profiles, center-to-annulus contrast, outer-ring contrast, footprint radius, eccentricity, compactness, local autocorrelation, nearest-neighbor distance, and overlap with other sites are computed from this map. A bounded specificity score is

$$S_i = \frac{A_i^{\text{ROI}} - A_i^{\text{ann}}}{|A_i^{\text{ROI}}| + |A_i^{\text{ann}}| + \epsilon}. \quad (7)$$

Matched spatial controls preserve local intensity, field region, distance from the image boundary, mask area, and, where possible, crowding. Controls include annulus-versus-outer-ring residuals and translated pseudo-ROIs that do not overlap protected labeled regions. Superiority, noninferiority, and equivalence are distinct hypotheses. Equivalence is claimed only if the complete confidence interval lies within a prespecified scientifically meaningful margin $[-\delta, \delta]$; failure to reject a difference is not described as equivalence (Lakens, 2017).

2.8. Scalar variance decomposition and observability

For each scalar metric m_{sb} observed at site s and burst b , the primary variance-component model is

$$m_{sb} = \beta_0 + u_s + v_b + \epsilon_{sb}, \quad u_s \sim \mathcal{N}(0, \sigma_s^2), \quad v_b \sim \mathcal{N}(0, \sigma_b^2). \quad (8)$$

With only four bursts, burst variance is interpreted cautiously and is accompanied by leave-one-burst-out and burst-fixed-effect analyses. The site-associated

intraclass correlation is

$$\text{ICC}_{\text{site}} = \frac{\sigma_s^2}{\sigma_s^2 + \sigma_b^2 + \sigma_\epsilon^2}. \quad (9)$$

Primary outcomes are Raw amplitude, residual amplitude, robust SNR, event area, time to peak, spatial specificity, footprint radius, annulus coupling, and known-positive recoverability. Cross-burst rank stability and bootstrap intervals provide nonparametric complements. Singular mixed models are downgraded to transparent fixed-effect or rank-based summaries rather than being reported as converged.

2.9. Functional and tensor decomposition

Functional analysis is conditional on trace-completeness, temporal-alignment, and held-out stability gates. Event-centered traces are modeled as

$$Y_{sb}(\tau) = \mu(\tau) + U_s(\tau) + V_b(\tau) + E_{sb}(\tau), \quad (10)$$

using a compact basis or functional principal components with simultaneous uncertainty bands. Functional mixed models follow established approaches for decomposing structured curve variation ([Morris and Carroll, 2006](#)).

A complementary low-rank model treats the site-by-burst-by-time array as a tensor:

$$\mathcal{X}_{sb\tau} \approx \sum_{k=1}^K a_{sk} c_{bk} h_{\tau k}. \quad (11)$$

Rank and stability are selected without using the held-out burst. A one-component solution would support a shared waveform modulated by site and burst gains; additional components are promoted only if they reproduce under leave-one-burst-out analysis and improve held-out reconstruction beyond a preregistered threshold. Tensor factors are descriptive, not automatically biological sources ([Kolda and Bader, 2009](#)).

2.10. Measurement phenotypes and known-positive recoverability

A *measurement phenotype* is defined as a reproducible profile of observability, not a biological cell type. Candidate dimensions include amplitude, SNR, repeatability, annulus coupling, spatial compactness, crowding, saturation, timing stability, and candidate rank. Continuous analyses are primary. Exploratory clustering is permitted only with bootstrap stability and without biological naming.

Known-positive recovery at fixed candidate budget is modeled with clustered or mixed-effects logistic regression:

$$\text{logit Pr}(M_{sb} = 1) = \beta_0 + \boldsymbol{\beta}^\top \mathbf{z}_{sb} + u_s + v_b, \quad (12)$$

where M_{sb} indicates whether a candidate matched the labeled occurrence and $\mathbf{z}_{sb}$ contains prespecified measurement covariates. Failures are partitioned into proposal, ranking, localization, timing, and non-maximum-suppression failures.

2.11. Positive-unlabeled evaluation and bounded-field annotation

Outside an exhaustively reviewed region, reported detection quantities are restricted to known-positive recall, proposal-union coverage, candidate burden, nearest-candidate rank, localization error, and failure category. Unmatched candidates remain unlabeled. No full-field precision, specificity, or false-positive rate is calculated.

A bounded spatial-temporal field is prepared for exhaustive review by two independent raters blinded to algorithmic candidates. Review categories are confirmed neuron, probable neuron, activity with uncertain identity, artifact, background, and unresolved. A third reviewer adjudicates disagreements. Inter-rater agreement is reported with raw agreement and category-appropriate chance-corrected statistics. Only this bounded field supports local precision-recall analysis, and confidence-sensitive analyses are reported alongside binary adjudication.

2.12. Frozen representation panel

The confirmatory representation panel is compact and interpretable: Raw or quiet-standardized carrier, signed temporal difference, frozen two-frame ICA as an equivalence control, ROI-minus-annulus residual, local coherence, lagged recurrence, and one prespecified information-divergence statistic. Parameters, candidate proposals, non-maximum suppression, and candidate budgets are frozen before confirmatory evaluation. The same proposals are reused where necessary to distinguish proposal gains from ranking gains. Broad feature expansion is prohibited.

2.13. Uncertainty, multiplicity, and sensitivity analyses

Primary estimands are declared before the protected rerun. Uncertainty is estimated by site-level, burst-level, or two-way resampling, preserving

temporal blocks. Exact or Monte Carlo sign-flip/permutation procedures are used when their exchangeability conditions are satisfied. Secondary families are controlled using the Benjamini–Hochberg false-discovery-rate procedure (Benjamini and Hochberg, 1995). Every primary analysis is repeated with burst 2 excluded; identity-sensitive analyses are repeated under separate-site and approved-merge views. Effect sizes and compatibility intervals are reported even when hypothesis tests are inconclusive.

2.14. Software and reproducibility

The analysis is implemented in the maintained **neurobench** package. Each stage records source hashes, label hashes, configuration, software environment, random seeds, output checksums, validation status, and exact figure/table generation commands. Existing scientific outputs are never overwritten. CPU-compatible tests are mandatory; CUDA may accelerate array operations but cannot change the numerical contract. The final result-to-manuscript exporter writes each numerical claim as a $\text{\LaTeX}$ macro and records its exact source in a machine-readable map.

3. Results

3.1. Current evidence status and protected-rerun requirement

The repository audit identified 79[†] labeled occurrences across 4[†] bursts. The current table contains 27[†] distinct coordinate pairs but only 26[†] proposed canonical identities. A provisional ROI 015-to-ROI 010 identity mapping is combined with geometry dictionaries keyed only by canonical identity in two signature-analysis paths, allowing one spatial site to overwrite the other. All identity-sensitive signature statistics are therefore considered provisional until trace extraction is rebuilt around immutable site identifiers. Table 1 summarizes the current evidence and the interpretation permitted before the rerun.

Table 1: Current repository evidence and the strongest interpretation permitted before the protected rerun. Daggered values are provisional.

Question	Current evidence	Permitted interpretation
Do labeled times contain unusual temporal activity?	ICA and signed difference both exceed shifted-label controls; empirical $p = 0.005^\dagger$.	Labeled windows contain transient change relative to the implemented temporal null.
Does two-frame ICA identify a novel operator?	$R^2 = 0.9938^\dagger$ against signed difference; event-score correlation $= 0.9988^\dagger$.	No independent added operation is established; the component is effectively derivative-like.
Is there a repeatable event-aligned Raw waveform?	Held-out correlation $= 0.8256^\dagger$; first temporal mode $= 87.4\%^\dagger$.	A strong recording/population-level burst waveform is present.
Is the Raw waveform neuron-specific?	Event-minus-annulus correlation difference $= -0.0131^\dagger$ (95% interval $-0.0277^\dagger$ to $0.0008^\dagger$).	No practically meaningful center advantage has yet been established at the tested support.
Does local subtraction retain event signal?	Residual peak-amplitude difference $= 91.33^\dagger$ (95% interval $50.64^\dagger$ to $140.75^\dagger$).	Event-associated localized amplitude remains after local subtraction.
Is residual morphology universal?	Residual-minus-spatial-control difference $= -0.0953^\dagger$ (95% interval $-0.2560^\dagger$ to $0.0472^\dagger$).	No universal neuron-specific residual waveform is established.
Do contextual features assist recovery?	Budget-20 known-positive recall: carrier $0.5409^\dagger$, coherence $0.6053^\dagger$, lagged recurrence $0.6220^\dagger$.	Context may improve proposal/ranking of labeled positives; precision remains unidentified.
Can full-field precision be reported?	Positive labels are not exhaustive.	No; unmatched candidates remain unknown outside an exhaustive review field.

3.2. Two-frame ICA behaves as a temporal-difference operator

The learned two-frame independent component was almost perfectly explained by signed temporal difference, with provisional $R^2 = 0.9938^\dagger$ and

event-score correlation $r = 0.9988^\dagger$. Both representations exceeded the available shifted-label controls with an empirical upper-tail probability of $0.005^\dagger$. The result establishes that the labeled intervals contain unusual temporal change under the current control construction, but it does not establish recovery of an independent neuronal source. Instead, the learned component is interpreted as an onset- or change-sensitive operator whose fluorescence fidelity must be assessed separately.

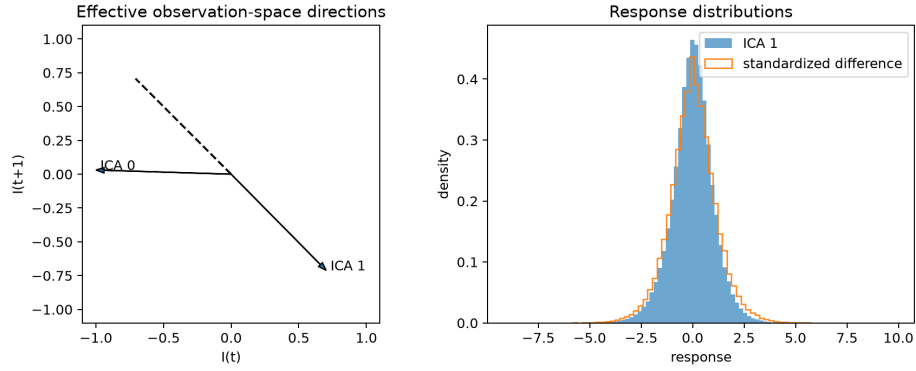


Figure 1: Operator-identification analysis for the learned two-frame representation. The protected rerun retains this comparison as an analytic equivalence control rather than treating ICA as an automatically novel representation. **Provisional:** this artifact predates the identity/geometry repair and is included only to guide the confirmatory rerun.

3.3. Raw event-aligned traces contain a strong shared waveform

The preliminary held-out Raw event-template correlation was $0.8256^\dagger$, and the first temporal principal component accounted for $87.4\%^\dagger$ of the event-aligned variance. These results are consistent with a strong shared burst waveform. However, the mean event-minus-matched-annulus correlation difference was $-0.0131^\dagger$ with a provisional 95% interval from $-0.0277^\dagger$ to $0.0008^\dagger$. Thus, the same morphology was not demonstrably stronger at the labeled center than in adjacent tissue at the tested support. The appropriate current conclusion is that a reproducible recording- or neighborhood-level event waveform exists; its neuron specificity has not been established.

3.4. Local subtraction reveals event-associated amplitude but not a universal residual shape

Subtracting a local annulus from the ROI increased event-associated residual peak amplitude relative to matched shifted windows by $91.33^\dagger$ intensity

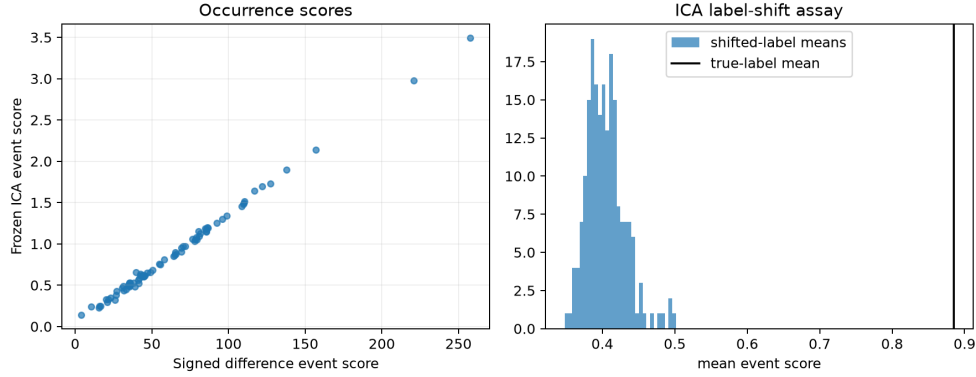


Figure 2: Preliminary external assay comparing Raw and learned/difference representations at labeled occurrences and shifted-time controls. **Provisional:** this artifact predates the identity/geometry repair and is included only to guide the confirmatory rerun.

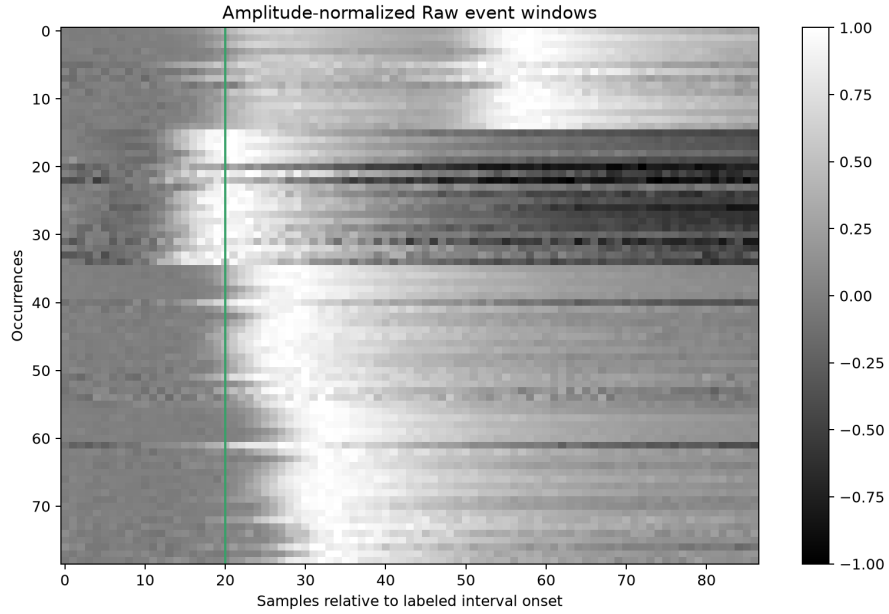


Figure 3: Preliminary event-aligned Raw trace heatmap. The final version will preserve immutable site identity, display native-amplitude and robustly standardized views separately, and annotate burst and field membership. **Provisional:** this artifact predates the identity/geometry repair and is included only to guide the confirmatory rerun.

units, with a provisional 95% interval from $50.64^\dagger$ to $140.75^\dagger$. This supports the presence of spatially localized event energy. In contrast, the held-out residual-template correlation was $0.5307^\dagger$, and the residual-minus-spatial-control correlation difference was $-0.0953^\dagger$ with a provisional interval from $-0.2560^\dagger$ to $0.0472^\dagger$. The current evidence therefore distinguishes localized amplitude from universal morphology: local signal remains after common-mode subtraction, but its shape is heterogeneous and does not yet outperform the matched spatial control.

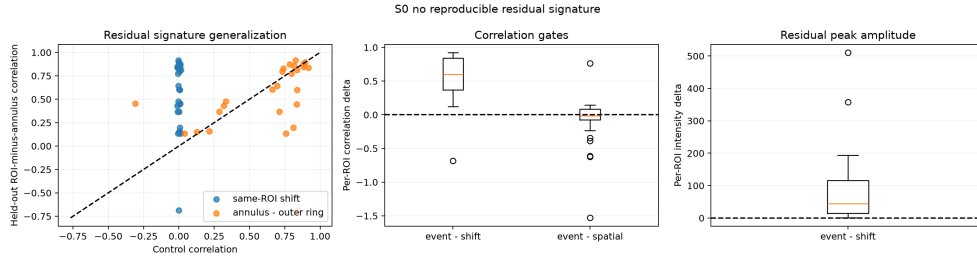


Figure 4: Preliminary ROI-minus-annulus residual-signature analysis. Final inference will be regenerated after the identity/geometry repair and will include multiple spatial supports, matched translated controls, and prespecified equivalence margins. **Provisional:** this artifact predates the identity/geometry repair and is included only to guide the confirmatory rerun.

3.5. Repeatability appears to reside more strongly in observability than waveform morphology

The preliminary median within-identity pairwise correlation was $0.3720^\dagger$ for Raw traces and $0.3144^\dagger$ for residual traces. These modest values do not support a universal neuron-specific waveform under the current normalization. Exploratory inspection of occurrence-level metrics nevertheless suggested that the rank ordering of event amplitude may be substantially more stable across bursts than detailed trace shape. This observation motivates the protected variance-decomposition analysis of amplitude, SNR, event area, timing, annulus coupling, and spatial compactness. No numerical intraclass-correlation estimate is promoted in the present draft because the current identity mapping can contaminate site-level aggregation.

Planned figure

A neuron-by-burst matrix of native Raw and residual amplitudes; cross-burst rank plots; variance components and site-associated ICCs; sensitivity to burst 2 and ROI 010/015 identity.

Figure 5: Stable site-dependent observability. The figure will test whether repeated neurons or observation sites possess stable amplitude, SNR, timing, or spatial-specificity profiles even when waveform shape is heterogeneous.

3.6. *Spatial/contextual features improve known-positive ranking without identifying precision*

In existing full-field screens, carrier-only known-positive recall at candidate budget 20 was $0.5409^\dagger$, compared with $0.6053^\dagger$ for a local-coherence lane and $0.6220^\dagger$ for a lagged-recurrence lane. These values indicate that spatial and temporal context can improve the ordering or proposal coverage of known positives. They do not identify full-field precision because unmatched candidates may be unlabeled neurons. The protected confirmation will hold proposals and non-maximum suppression fixed where possible, separate proposal from ranking gains, and reserve precision estimation for the exhaustively reviewed bounded field.

Planned figure

Recall-versus-candidate-budget curves for the frozen representation panel, proposal-union ceilings, same-proposal ranking comparisons, and failure categories. A separate panel will report bounded-field precision–recall after blinded exhaustive review.

Figure 6: Detection implications under positive–unlabeled annotation.

3.7. Acquisition structure is a potential determinant of observability

Preliminary quiet-frame analysis reported an intensity–variance slope of approximately $3.63^\dagger$ with weighted R^2 between $0.955\text{--}0.959^\dagger$. The audit also identified a provisional field boundary near $x = 286^\dagger$ and a left–right global-trace correlation of $0.557^\dagger$. These observations motivate stratification by field, saturation, baseline intensity, motion, and local effective rank. The final analysis will determine whether recurrently difficult sites are explained by biological event magnitude, acquisition position, local contamination, or a combination of these factors.

Planned figure

Variance-versus-intensity fit, saturation and motion maps, field-boundary diagnostics, temporal spectra, and a comparison of the canonical recording with any available unlabeled recordings.

Figure 7: Acquisition physics and recording-quality context.

3.8. Prespecified confirmatory hypotheses

The protected rerun follows the hypothesis hierarchy in Table 2. Functional and tensor analyses are promoted only if trace completeness, held-out reconstruction, and component-stability gates pass. A null or heterogeneous result remains scientifically reportable because it defines the boundary between shared burst detection and neuron-specific identification.

Table 2: Prespecified hypothesis hierarchy for the protected analysis.

ID	Hypothesis	Current status	Confirmatory decision rule
H1	Annotated events differ from matched ordinary times.	Supported provisionally.	Site/burst-aware block-shift effect and interval exclude the null region.

Continued on next page

Table 2 continued from previous page

ID	Hypothesis	Current status	Confirmatory decision rule
H2	Event activity is spatially concentrated at the labeled site.	Partial amplitude evidence.	ROI contrasts outperform matched spatial controls; equivalence assessed only with frozen margins.
H3	Sites or neurons possess stable activation gain/observability.	Strong exploratory motivation.	Cross-burst rank stability and site variance component pass preregistered reliability gates.
H4	Neurons possess reproducible neuron-specific temporal waveforms.	Not established.	Functional site component improves held-out reconstruction and exceeds spatial controls.
H5	Measurement phenotype explains known-positive recovery failures.	Suggestive.	Prespecified covariates improve held-out recovery model with stable coefficient direction.
H6	Two-frame ICA adds information beyond temporal difference.	Provisionally rejected.	Promote only if repaired rerun breaks the analytic equivalence and adds utility/fidelity.
H7	Compact spatial/contextual statistics add recovery information.	Supported provisionally.	Frozen same-proposal and proposal-union comparisons improve known-positive recall without disproportionate burden.
H8	Directed or network interactions are identifiable.	Not established.	Exploratory only; no causal promotion from this recording.

4. Discussion

4.1. Change detection is not source identification

The central methodological distinction in this study is between detecting temporal change and identifying a neuron-specific source. A two-frame operator can respond strongly at the onset of a burst while discarding slow fluorescence morphology or preserving activity that is equally visible in surrounding tissue. The near-equivalence between the learned ICA component and signed difference is therefore not merely a negative result. It identifies the operation that the learning procedure selected under this data geometry and

prevents an unsupported interpretation of the component as an independently recovered neuronal signal.

This distinction is consistent with the modular structure of established calcium-imaging methods, which separately model spatial footprints, temporal traces, background activity, and deconvolved events (Pnevmatikakis et al., 2016; Giovannucci et al., 2019; Zhou et al., 2018). The appropriate comparison for NeuRev is not whether every representation resembles one of these packages, but whether each proposed operator contributes a measurable function: temporal-change sensitivity, spatial localization, nuisance removal, waveform preservation, or candidate ranking.

4.2. A shared burst waveform can coexist with local neuronal information

The current evidence supports a strong event-aligned Raw waveform while showing that nearby tissue contains a similar morphology. This pattern is compatible with several mechanisms, including widespread neural recruitment, neuropil contamination, motion-coupled fluorescence, common illumination or acquisition dynamics, or spatially broad indicator responses. The data do not yet identify their relative contributions. Local annulus subtraction removes part of the shared component and leaves event-associated amplitude, but it can also suppress real spatially broad neural signal. Consequently, residualization should be evaluated as a measurement transformation, not assumed to yield ground truth.

The proposed functional and tensor decompositions make this ambiguity explicit. A dominant rank-one structure would support a shared temporal waveform modulated by site and burst gains. Reproducible additional components could indicate kinetic or spatial substructure, but they would still require matched spatial controls and held-out stability before being interpreted as neuron-specific. Conversely, failure to find stable additional components would bound the amount of neuron-specific waveform information identifiable from this recording.

4.3. Stable observability may be the most productive neuron-level target

Detailed activation morphology need not be the only reproducible property of a neuron. A site may repeatedly exhibit high amplitude, strong SNR, compact spatial support, weak annulus coupling, or reliable candidate ranking even when its exact waveform varies. We refer to this profile as a measurement or observability phenotype. The term is deliberately nonbiological: stable

visibility can arise from expression level, optical depth, focus, cell size, local crowding, acquisition field, or true response magnitude.

Quantifying observability has direct methodological value. It can explain why some known neurons are systematically recovered while others are repeatedly missed, reveal whether apparent algorithm failures are actually proposal or acquisition failures, and guide the next annotation or data-collection effort. A detector that improves only for already bright, isolated neurons is less scientifically useful than one that expands coverage of difficult but credible sites. The proposed variance-component and recoverability models therefore treat site-associated heterogeneity as a primary result rather than nuisance variation.

4.4. Incomplete annotation changes what detection metrics mean

Sparse positive labels make recall of known observations identifiable but leave full-field precision unidentified. Treating every unmatched candidate as background would penalize the algorithm for discovering neurons omitted by the annotation process and would favor conservative methods that reproduce the labeling bias. Positive-unlabeled learning theory emphasizes that conclusions depend on how positives enter the labeled set, particularly when labeling is not random (Bekker and Davis, 2020).

The bounded-field annotation study is designed to convert one small region from positive-unlabeled to exhaustively reviewed data. Its value is disproportionate to its size: it permits a local precision estimate, a false-candidate taxonomy, an estimate of hidden-positive prevalence, and an assessment of whether candidate score calibrates review confidence. The rest of the full field remains useful for known-positive recovery and discovery, but it should not be assigned a fictitious negative class.

4.5. Implications for unsupervised learning in NeuRev

Avoiding large supervised models is appropriate given the current data volume, but unsupervised learning does not remove the need for scientific evaluation. An unsupervised representation can learn acquisition structure, global burst timing, local covariance, or a simple derivative. The relevant question is whether it exposes a stable and useful statistic that an interpretable comparator does not already provide.

The confirmatory program therefore places learned and analytic operators in the same frozen panel. Pairwise ICA remains as an equivalence control; local coherence and lagged recurrence are tested for ranking utility; residualization

is tested for spatial specificity and fidelity; and Raw remains the carrier against which information loss is measured. New learning is warranted only when the existing analysis identifies a missing operation or a reproducible residual structure that cannot be represented by the compact panel.

4.6. Limitations

The study has several non-negotiable limitations. First, one recording cannot establish generalization across animals, preparations, sessions, or microscopes. Second, four shared bursts provide limited independent temporal replication and make burst-level variance estimates imprecise. Third, current labels are sparse and partly uncertain, and the same bursts have informed multiple exploratory decisions. Fourth, a provisional identity merge exposed a geometry-indexing bug, so pre-repair identity-sensitive numbers cannot be final. Fifth, annulus subtraction is not a pure nuisance-removal operation. Sixth, candidate-based metrics outside the bounded field cannot identify precision or specificity. Seventh, any coactivity or lag analysis is descriptive because shared drive and calcium kinetics can generate apparent network structure.

These limitations do not make the dataset scientifically unusable. They determine the level at which it should be analyzed. The strongest contribution is a transparent within-recording measurement study, accompanied by explicit criteria for the additional data and annotation required to promote each claim.

4.7. Highest-value next evidence

The most valuable immediate addition is not a wider hyperparameter sweep. It is a repaired site-level trace atlas, final adjudication of identity and timing questions, and exhaustive review of one bounded field. The most valuable future dataset is an independently acquired recording that satisfies a frozen quality and annotation contract. Such a recording would permit true recording-level external validation of the shared-versus-specific decomposition and the frozen representation panel.

5. Conclusion

Sparse zebrafish fluorescence data require a distinction between event detection, spatial localization, source identification, and biological interpretation. The current NeuRev evidence is most coherent when framed as a measurement

Table 3: Limitations and the corresponding manuscript safeguard.

Limitation	Safeguard
One canonical recording	Restrict claims to within-recording measurement structure; require an independent recording for external generalization.
Four shared burst intervals	Use leave-one-burst-out and burst-exclusion sensitivity; avoid asymptotic burst-level claims.
Sparse, partly uncertain positives	Use positive-unlabeled metrics and preserve confidence/provenance; estimate precision only in a bounded exhaustive field.
Identity/geometry collision in existing assays	Repair before trace aggregation; preserve original sites and regenerate all identity-sensitive results.
Repeated exploratory use of the same bursts	Freeze the compact panel and estimands before a protected confirmatory rerun.
Local annulus may contain true neural signal	Report Raw, annulus and residual jointly; treat subtraction as a transformation, not ground truth.
Acquisition heterogeneity and saturation	Characterize noise, field effects, motion and saturation before interpreting stable observability.
Shared drive can induce network correlations	Keep connectivity analyses descriptive and condition on global/local common mode.

and identifiability study: temporal differentiation detects burst-associated change; Raw traces contain a strong shared waveform; local subtraction retains event-associated amplitude; and neuron-specific morphology remains heterogeneous. The protected analysis will determine whether stable site-dependent observability provides a more reproducible neuronal signature than waveform shape. By preserving spatial-site identity, respecting positive–unlabeled labels, separating detection utility from trace fidelity, and reserving precision for exhaustively reviewed data, the proposed framework converts a data-limited detection problem into a sequence of falsifiable scientific questions.

Declaration of competing interests

[AUTHOR DECISION: Confirm that the authors have no competing financial or personal interests, or provide the required disclosure.]

Ethics statement

[AUTHOR DECISION: Insert the animal-use protocol, approving institution, protocol number, and confirmation that procedures followed applicable guidelines. Do not infer these details from the repository.]

Funding

[AUTHOR DECISION: List funding sources, grant numbers, and the role of each funder.]

CRedit authorship contribution statement

[AUTHOR DECISION: Complete CRediT roles for every author. Suggested roles to review include Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Visualization, Writing—original draft, Writing—review & editing, Supervision, Project administration, and Funding acquisition.]

Data availability

The release manuscript will cite a versioned data and annotation package or state the access restrictions and their justification. At minimum, the package will include immutable observation identifiers, site geometry, canonical-identity decisions, burst intervals, annotation provenance, configuration files, source hashes, generated result macros, and a figure/table source map. [AUTHOR DECISION: Select public repository/accession, controlled-access process, or justified non-release statement.]

Code availability

The analysis code will be released from a versioned NeuRev Workbench revision together with an executable environment specification and the protected-run configuration. [AUTHOR DECISION: Insert repository URL, release tag, and archival DOI when available.]

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Declaration of generative AI and AI-assisted technologies in the writing process

During manuscript preparation, the authors used OpenAI language-model tools to help organize the analysis plan and draft a manuscript skeleton. The authors are responsible for verifying all text, code, citations, numerical results, and interpretations and will review and edit the final submission. [AUTHOR DECISION: Revise this disclosure to match the journal’s submission form and the tools actually used in the final manuscript.]

Appendix A. Manuscript result-status convention

Values marked with a dagger are imported from pre-repair repository artifacts and are provisional. They must be regenerated after the observation-site/canonical-identity repair. The final manuscript build must fail if a provisional macro remains enabled.

References

- Ahrens, M.B., Orger, M.B., Robson, D.N., Li, J.M., Keller, P.J., 2013. Whole-brain functional imaging at cellular resolution using light-sheet microscopy. *Nature Methods* 10, 413–420. doi:[10.1038/nmeth.2434](https://doi.org/10.1038/nmeth.2434).
- Bekker, J., Davis, J., 2020. Learning from positive and unlabeled data: A survey. *Machine Learning* 109, 719–760. doi:[10.1007/s10994-020-05877-5](https://doi.org/10.1007/s10994-020-05877-5).
- Benjamini, Y., Hochberg, Y., 1995. Controlling the false discovery rate: A practical and powerful approach to multiple testing. *Journal of the Royal Statistical Society: Series B (Methodological)* 57, 289–300. doi:[10.1111/j.2517-6161.1995.tb02031.x](https://doi.org/10.1111/j.2517-6161.1995.tb02031.x).
- Foi, A., Trimeche, M., Katkovnik, V., Egiazarian, K., 2008. Practical poissonian–gaussian noise modeling and fitting for single-image raw-data. *IEEE Transactions on Image Processing* 17, 1737–1754. doi:[10.1109/TIP.2008.2001399](https://doi.org/10.1109/TIP.2008.2001399).
- Friedrich, J., Zhou, P., Paninski, L., 2017. Fast online deconvolution of calcium imaging data. *PLOS Computational Biology* 13, e1005423. doi:[10.1371/journal.pcbi.1005423](https://doi.org/10.1371/journal.pcbi.1005423).
- Giovannucci, A., Friedrich, J., Gunn, P., Kalfon, J., Brown, B.L., Koay, S.A., Taxidis, J., Najafi, F., Gauthier, J.L., Zhou, P., Khakh, B.S., Tank, D.W., Chklovskii, D.B., Pnevmatikakis, E.A., 2019. CaImAn: An open source tool for scalable calcium imaging data analysis. *eLife* 8, e38173. doi:[10.7554/eLife.38173](https://doi.org/10.7554/eLife.38173).
- Keemink, S.W., Lowe, S.C., Pakan, J.M.P., Dylid, E., van Rossum, M.C.W., Rochefort, N.L., 2018. FISSA: A neuropil decontamination toolbox for calcium imaging signals. *Scientific Reports* 8, 3493. doi:[10.1038/s41598-018-21640-2](https://doi.org/10.1038/s41598-018-21640-2).
- Kolda, T.G., Bader, B.W., 2009. Tensor decompositions and applications. *SIAM Review* 51, 455–500. doi:[10.1137/07070111X](https://doi.org/10.1137/07070111X).
- Lakens, D., 2017. Equivalence tests: A practical primer for t tests, correlations, and meta-analyses. *Social Psychological and Personality Science* 8, 355–362. doi:[10.1177/1948550617697177](https://doi.org/10.1177/1948550617697177).

- Morris, J.S., Carroll, R.J., 2006. Wavelet-based functional mixed models. *Journal of the Royal Statistical Society: Series B (Statistical Methodology)* 68, 179–199. doi:[10.1111/j.1467-9868.2006.00539.x](https://doi.org/10.1111/j.1467-9868.2006.00539.x).
- Pnevmatikakis, E.A., Soudry, D., Gao, Y., Machado, T.A., Merel, J., Pfau, D., Reardon, T., Mu, Y., Lacefield, C., Yang, W., Ahrens, M., Bruno, R., Jessell, T.M., Peterka, D.S., Yuste, R., Paninski, L., 2016. Simultaneous denoising, deconvolution, and demixing of calcium imaging data. *Neuron* 89, 285–299. doi:[10.1016/j.neuron.2015.11.037](https://doi.org/10.1016/j.neuron.2015.11.037).
- Zhou, P., Resendez, S.L., Rodriguez-Romaguera, J., Jimenez, J.C., Neufeld, S.Q., Giovannucci, A., Friedrich, J., Pnevmatikakis, E.A., Stuber, G.D., Hen, R., Kheirbek, M.A., Sabatini, B.L., Kass, R.E., Paninski, L., 2018. Efficient and accurate extraction of in vivo calcium signals from microendoscopic video data. *eLife* 7, e28728. doi:[10.7554/eLife.28728](https://doi.org/10.7554/eLife.28728).